

DEMAND ROSTERING DATA ANALYST

Focused Interview Preparation

Power BI + Forecasting + Feature Store + Data Investigation

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Use one Coles story throughout:

Forecasting models consume curated features from the Feature Store, and Power BI applications sit on top of forecasting outputs and related data. Use this flow for technical, analytical, STAR and stakeholder questions.

1. What to expect

Expect direct technical questions, data-investigation scenarios and STAR questions. The central test is whether you can move from a Power BI number back through the reporting layer, forecast output, Feature Store data and source/business rules. Do not prepare unrelated Coles project stories.

Question type	What they test	Your anchor
STAR	What you actually did	Power BI / forecasting / Feature Store
Scenario	How you investigate	Report -> forecast -> features -> source/rules
SQL / modelling	Grain, joins, reconciliation	Forecast vs actual
Power BI	Semantic model, DAX, troubleshooting	Apps on forecasting data
Fabric	Transferable platform skills	Lakehouse/notebook patterns

2. The one architecture story

At Coles, forecasting use cases consume engineered and curated features from the Feature Store. Forecast outputs and related business data are exposed to users through Power BI applications. When a number looks wrong, the investigation is end-to-end rather than stopping at the visual.

Power BI app / visual -> semantic model & measure -> forecast output -> Feature Store features -> upstream source data & business rules

For demand rostering, translate this to venue demand. The domain changes, but the pattern remains: forecast, actual, operational measures, Power BI, data quality and discrepancy investigation.

3. 60-second introduction

Tell me about yourself.

I currently work in Coles Data & AI, where a major part of my work supports forecasting use cases. We have a Feature Store that prepares reusable, curated features for forecasting models, and Power BI applications sit on top of the forecasting outputs and related data for business consumption. My role is end-to-end: understanding requirements and business rules, designing the data solution, validating the data, supporting production and investigating issues when downstream numbers do not look right. That is why this role is relevant to me. Although the business problem here is demand rostering, the core work - forecast versus actual analysis, data modelling, Power BI, data quality and tracing discrepancies through data layers - is very close to the problems I already work with.

4. STAR questions - highest priority

Use STAR naturally. Keep Situation and Task short. Spend most of the answer on Action and Result.

STAR 1 - Tell me about a time the business said a Power BI number was wrong.

Situation/Task: A business-facing Power BI application showed a value that did not align with the expected forecasting result, and I needed to establish where the discrepancy originated. Action: I reproduced the issue using the same filters and grain, checked the semantic measure and aggregation, and reconciled the report value to the forecast output. I then traced the affected keys to the relevant Feature Store features and upstream business rules, comparing record counts, dates, keys and expected values at each boundary. After identifying the owning layer, I validated the correction all the way back through Power BI and explained the cause in business terms. Result: The visible discrepancy was resolved and validation was strengthened so the same type of issue could be detected earlier.

STAR 2 - Tell me about a difficult data-quality issue.

Use a genuine forecasting/Feature Store issue. State the exact symptom and affected grain. Explain how you checked completeness, duplicates, mappings, date logic, joins and transformation rules, then reconciled the corrected feature through the forecast output and Power BI. The point is to demonstrate disciplined investigation, not to introduce another project.

STAR 3 - Tell me about translating an ambiguous business requirement.

Use a forecasting requirement behind a Power BI application. Clarify the decision the business is trying to make, metric definition, grain, historical/future period and business rules. Map those requirements to Feature Store data and downstream forecast/reporting needs, then use examples and playback with stakeholders before implementation.

STAR 4 - Tell me about working with technical and non-technical stakeholders.

Use the same Power BI forecasting flow. A business user may say only that a number looks wrong. Convert that into an exact venue/location, date, metric, filter and expected result; work with the relevant technical owners using keys and lineage; then translate the root cause and impact back into business language.

STAR 5 - Tell me about a production issue you owned end-to-end.

Choose one genuine issue from the forecasting/Feature Store-to-Power-BI flow that you remember clearly. Structure it as detection -> impact -> reproduce -> trace -> fix -> validate -> communicate -> prevent. Do not introduce Range Optimiser or migration examples unless the interviewer explicitly asks for another project.

5. Scenario questions - very likely

A venue manager says the demand forecast shown in Power BI is wrong. What do you do?

First make the complaint specific: venue, date, time interval, metric, expected value and comparison point. Reproduce the Power BI result with the same filters, check the measure and semantic-model relationships, and reconcile it to the forecast output. If reporting is correct, inspect the relevant forecast inputs/features at the same grain and trace them back to source data/business rules. Compare forecast against actuals and determine whether the issue is isolated or systematic. Then classify it as reporting, data, business-rule or forecast-performance issue.

Power BI shows 12,000 but SQL gives 10,000.

Align filters, date boundaries, grain and metric definition. Check DAX aggregation, relationships, row multiplication, refresh/version differences and whether SQL is querying the same curated dataset. Reconcile progressively instead of changing the report until you know where the divergence begins.

Forecast is consistently higher than actual demand.

Quantify bias by venue, weekday and interval. Confirm forecast and actual use the same business date and grain. Inspect relevant input features for stale, missing or incorrectly mapped values. If the data is valid, provide the pattern and evidence to the forecasting/model owner rather than calling it a data defect.

Only one venue is wrong while all others look correct.

Start with venue-specific mappings and data rather than a global calculation. Compare the affected venue with a known-good venue through Power BI, forecast output and Feature Store keys, looking for mapping, missing-data, date/calendar or source differences.

Yesterday's dashboard changed after today's refresh.

Check forecast version/as-of timestamp, late-arriving actuals, backfills and Feature Store corrections. Clarify whether the report should show the latest forecast or the forecast that existed at the original operational cutoff.

6. Power BI questions

How do you troubleshoot a wrong Power BI number?

Visual filters -> measure definition -> relationships/filter propagation -> underlying forecast/actual data at the same grain -> upstream Feature Store. Do not assume Power BI is the cause.

Measure vs calculated column?

A calculated column is computed row by row and stored. A measure is evaluated at query time under filter context. KPIs such as forecast variance normally belong in measures.

What is filter context?

The filters coming from slicers, axes, page/report filters and relationships. The same measure can return different results as filter context changes.

What does CALCULATE do?

It evaluates an expression under modified filter context. Explain it using forecast measures filtered to a period, location group or comparison window.

Why a star schema?

Clear fact grain, reusable dimensions, predictable relationships and simpler DAX. For rostering, think Date, Venue and Time/Role dimensions around forecast, actual and roster facts.

How do you avoid double counting?

Declare grain, validate relationship cardinality, avoid combining incompatible fact grains directly, and reconcile totals before and after joins/relationships.

How do you validate a Power BI release?

Reconcile key measures to controlled SQL/source cases, test filters and drill paths, compare totals at several grains, test edge cases and confirm refresh/version timestamps.

7. SQL / data questions

How do you compare forecast and actual?

Aggregate both to the same agreed grain before joining. Keep unmatched records visible for reconciliation. Calculate absolute error, percentage error and bias, handling zero actuals explicitly.

How do you find duplicates?

GROUP BY the declared business key and HAVING COUNT(*) > 1. The key must reflect the real grain and forecast version where relevant.

Why can a join inflate totals?

One or both sides are not unique at the join key. Profile uniqueness and compare row counts and sums before and after the join.

How do you get the latest forecast?

Use ROW_NUMBER over the business key ordered by forecast generation timestamp/version descending, retaining row_number = 1 only when 'latest' is the required business definition.

How do you identify missing periods?

Build an expected date/time spine for valid operating periods, left join forecast/actual data and identify missing keys.

What data-quality checks matter?

Completeness, uniqueness, valid ranges, referential integrity/mappings, freshness, expected coverage by grain and reconciliation between Feature Store, forecast output and reporting layer.

8. Translate your Coles experience to demand rostering

Your Coles context	Demand rostering equivalent
Location-level forecasting	Venue-level demand forecasting
Forecast output consumed in Power BI	Demand forecast consumed in Power BI
Feature Store inputs	Demand drivers/features
Forecast vs actual validation	Forecast demand vs actual demand
Business-facing Power BI application	Operations/rostering dashboard

Forecast error vs rostering error?

Forecast error is predicted demand versus actual demand. Rostering error is required labour versus labour rostered. A good forecast can still produce a poor roster if staffing rules or roster decisions are wrong; a roster can also look wrong because the forecast was wrong.

What grain would you expect?

Confirm it rather than assume it. Venue + business date + time interval is a likely analytical grain, potentially extended by department or role. Forecast, actual and labour measures must be reconciled at compatible grains.

9. Fabric - keep it honest and short

What is your Fabric experience?

My production experience is stronger on Azure Databricks, but the relevant patterns are transferable: lakehouse data, Delta, SQL/Python/Spark transformations, notebooks, curated analytical models, data quality and production pipelines. For this role I would learn the Fabric-specific implementation while applying those existing patterns. I would not present myself as a Fabric expert if I have not used it to that depth.

10. What NOT to add

- Do not introduce Range Optimiser unless they explicitly ask for another project.
- Do not introduce migration stories to answer a data-quality question.
- Do not explain internal Coles framework names or acronyms.
- Do not claim that you built the forecasting model if your ownership was the data/Feature Store and downstream integration.
- Do not present yourself as a Power BI specialist beyond your genuine hands-on depth.

11. Rapid question bank

- Tell me about yourself.
- Why this Demand Rostering role?
- Tell me about a time a Power BI number did not match expectations.
- Tell me about a difficult discrepancy you traced end-to-end.
- Tell me about clarifying an ambiguous metric or rule.
- Tell me about a production data issue you owned.
- Tell me about explaining a technical issue to a business stakeholder.
- How would you investigate forecast vs actual variance?
- How do you decide whether the issue is Power BI, data or model performance?
- How would you detect systematic forecast bias?
- How do you establish dataset grain?
- How can joins cause double counting?

- How do you reconcile two datasets safely?
- How do you handle missing/late data?
- What DQ controls belong around forecast data?
- Measure vs calculated column?
- Explain filter context and CALCULATE.
- How do you troubleshoot an incorrect Power BI visual?
- How would you model forecast, actual and roster data?
- What is your Fabric experience?
- How would notebooks fit?
- What is forecast error vs rostering error?
- How would you explain a discrepancy to a venue manager?
- How would you prove a forecasting issue is not a reporting issue?

12. Scenario answer framework

1. Clarify the business symptom and metric.
2. Confirm grain, date/time, filters and version.
3. Reproduce the Power BI result.
4. Check semantic model / DAX.
5. Reconcile to forecast output.
6. Trace relevant Feature Store features.
7. Trace source/business rule only where needed.
8. Quantify impact: isolated or systemic?
9. Fix at the owning layer and validate end-to-end.
10. Explain the result clearly and add prevention/monitoring.

13. Final preparation checklist

- One genuine Power BI/forecast discrepancy story with exact details.
- One genuine Feature Store data-quality story.
- One requirements/stakeholder story from the same forecasting ecosystem.
- Practise the 60-second introduction aloud.
- Practise report -> semantic model -> forecast -> Feature Store -> source without internal jargon.
- Revise DAX: measures, filter context, CALCULATE, DIVIDE and time comparisons.
- Revise SQL: grain, joins, GROUP BY, window functions, duplicates and reconciliation.
- Be precise about what you owned versus what the forecasting/model team owned.

Core message: I can trace an operational number from Power BI back through the forecasting data and Feature Store, understand why it differs, and explain the result to the business.